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Speech Emotion Recognition with Wav2Vec 2.0 for Healthcare Applications

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1 Introduction

Emotions determine how we see the world, and how we interpret the actions of others.

– Paul Ekman

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1.1 Speech Emotion Recognition

Human speech is one of the most powerful and sophisticated communication devices. Besides conveying verbal information, it transmits a huge number of emotional messages. By varying **tone, pitch, rhythm, and volume**, the voice can express our state of mind more accurately than posture or facial expression. This capacity to perceive and deduce emotions from auditory data is at the core of human communication, stimulating knowledge and compassion.

Paul Ekman, an early researcher of emotions, listed **six universal emotions** that are shared: **joy, sadness, fear, anger, surprise, and disgust** [1]. They are universally recognized across cultures and can be accessed through voice modulations, which makes the voice a privileged place of emotional recognition.

In the past several years, the increasing availability of vocal data and advancements in **artificial intelligence (AI)** has provided an opportunity for **speech emotion recognition (SER)** systems. Such systems aim to automate the recognition of emotions in human language using advanced **signal processing, machine learning, and deep neural networks**. **SER systems** analyze vocal traits such as **pitch, intensity, and spectral characteristics** so that emotional states can be determined with remarkable accuracy.

1.2 Objectives of the Thesis

The main aim of this thesis is to study and investigate the performance of SER models when trained with Italian data. For this purpose, the study involves:

- **Dataset Preparation:** Applying data selection methods to maintain a balanced dataset taking advantage of distinct Italian datasets.
- **Model Training:** Fine-tuning a Transformer model and conducting the final training process.
- **Performance Evaluation:** Analyzing results on a specified test set, additional external datasets, and on a benchmark to provide a comprehensive evaluation.
- **Cross-Corpus Validation:** Examining the performance drop observed during cross-corpus validation and exploration of possible explanations for such limitations.

By tackling such issues, the thesis tries to contribute insights into the generalization and robustness of SER models, especially under cross-corpus conditions.

1.3 Significance and Motivation

Recent research on **SER** has made strong presentations within controlled environments. Our organization, **GPI** [2], made an initial data release using the Talking About algorithm to detect postpartum depression by voice examination. The experiment not only stressed the value of existing SER procedures, but also found important drawbacks that led to further evolution. Inspired by this research, this thesis project has the mission of creating version 2.0 of the *Talking About* algorithm. The effort will be devoted to improving the model's **robustness** and **generalizability** in real-world environments. Specifically, this study addresses the serious issue of **performance degradation** observed in **cross-corpus validation** scenarios, where models with good performance in a specific dataset tend to perform badly in others.

To help address these issues, the research is focused on:

- **Bridging the gap** between experimental success and practical utility.
- Give a concise explanation of **performance variations** between various datasets.
- Suggest reasonable solutions to enhancing the **robustness** and **generalizability** of SER models.

2 Literature Review

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2.1 Overview of SER Techniques

In recent years, SER has progressed from traditional signal processing methods to current deep learning-based approaches and has significantly improved performance. Despite this, cross-corpus generalization, data imbalance, and adversarial environmental noise robustness continue to impact the performance of SER systems in real-world applications.

In this chapter, the overall background of procedures used in SER is presented, with a particular emphasis on three general categories: **data augmentation**, **feature extraction** and **classification models** [3]. The following section, presents how different data augmentation methods, i.e., noise injection, pitch shifting, and time changing, have been utilized to make the SER systems stronger and generalize better. The second section deals with the most popular speech features, and the emphasis is on **Mel-Frequency Cepstral Coefficients (MFCCs)**, which remain the basis for SER research [4]. The third section outlines conventional machine learning and deep learning approaches, quoting their advantages, disadvantages, and recent developments in the domain.

Finally, we discuss conventional machine learning strategies along with relatively novel deep approaches to learning, pointing out the advances that have propelled the field. Despite the advances, future work will have to deal with outstanding challenges, most particularly concerning low model generalizability across corpora and against noisy conditions. Additional investigation of data augmentation techniques and novel feature extraction procedures will be critical to maintaining the performance of the SER system under changing **real-world conditions**.

2.1.1 Data Augmentation

One of the primary barriers to SER is that few emotional speech corpora have already been annotated. Emotion speech annotation and gathering takes a long process and are expensive, and tend to cause data skewness with certain emotions being inadequately represented. To offset these, several approaches have been used [5], including:

- **Time-Based Transformations:**
 - **Time Stretching:** Alters the speed of the speech signal but not the pitch.
 - **Time Shifting:** Shifts the waveform forward or back in time.
- **Frequency-Based Transformations:**
 - **Pitch Shifting:** Changes the pitch but keeps the length the same.
 - **Vocal Tract Length Perturbation (VTLP):** Simulates changes in vocal tract length to enhance overall generalizability.
- **Reverb and noise injection:**
 - **Additive Gaussian Noise:** Injects known quantities of Gaussian noise to mirror real life.
 - **Room Impulse Response (RIR) Simulation:** Utilizes convolution with pre-recorded room impulse responses to model spaces with reverb.
- **Synthetic Data Generation:**
 - **GAN-Based Data Augmentation:** Utilizes the Generative Adversarial Networks (GANs) to produce synthetic emotional samples of speech [6].
 - **Voice Conversion Models:** Transforms neutral speech to different emotion states.

As data augmentation generalizes and prevents overfitting, it also introduces artefacts or distortions. Quality sets are ensured by the appropriate selection of methods, enhancing the robustness of SER models under noisy real-world conditions.

2.1.2 Feature Extraction Techniques

Feature extraction is a significant task in SER as it converts raw speech signals to easily interpretable representations that can be classified. Some of the popular types include:

- **Spectral Features:** These are features that extract frequency-related information and include:
 - **Mel-Frequency Cepstral Coefficients (MFCCs):** MFCCs model human hearing by mapping frequencies to a logarithmic **Mel scale**, which increases the accuracy of emotion recognition across various datasets [7].
 - **Linear Predictive Coding (LPC):** Forecasts the formant form of the vocal tract to simulate speech production.
- **Prosodic Features:** Highlighting the speech intonation and rhythm, including:
 - **Pitch (Fundamental Frequency, F0):** Refers to the speaker's vocal emotion and tone intensity.
 - **Energy and Loudness:** Higher energy tends to be associated with anger, whereas lower energy tends to be associated with sadness.
- **Temporal Features:** Represent speech dynamics across the duration, including speech rate and articulation rate.
 - **Delta MFCCs:** First-order time derivatives of MFCCs, representing the dynamic change of speech over time, for example, changing pitch or rhythm.
 - **Delta-Delta MFCCs:** Second-order time derivative of MFCCs, capturing even higher-order temporal evolvments in speech, representing the sudden change in affect states.
 - **Zero-Crossing Rate (ZCR):** Measures the change in the sign of the signal, potentially identifying emotion-related speech dynamic variation.

2.1.3 Machine Learning Approaches in SER

After extraction, the extracted features are fed to the classifiers for emotion recognition. The most widespread techniques are:

- **Traditional Machine Learning Models:**
 - **Support Vector Machines (SVMs):** Work well on little datasets due to the insensitivity to high-dimensional space.
 - **Random Forests & Decision Trees:** These are used for classification and feature importance computation.
 - **Hidden Markov Models (HMMs):** Traditionally dominant in SER but less dominant in deep learning models.
- **Deep Learning Models:**
 - **Convolutional Neural Networks (CNNs):** Extract high-level features from spectrograms, often outperforming traditional methods.
 - **Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) & GRUs:** They capture temporal dependencies in speech waves, which are essential for emotion modeling.
 - **Transformer-based Models:** Recent studies show that **self-attention mechanisms** can be utilized to enhance emotion recognition accuracy [8].

2.2 Related Work and Previous Studies

Research in recent years has explored the use of **data augmentation** as a way of mitigating data variability and imbalance problems in SER. Noise addition, time stretching, pitch shifting, and vocal tract length perturbation are some of the techniques used to artificially increase training sets. The techniques assist in enhancing the generalizability capacity of models by exposing them to more variations of speech samples that better represent real-world speech variability [9].

It has been demonstrated through studies that data augmentation can greatly enhance the performance of both conventional machine learning approaches and deep learning models, including CNNs and LSTMs, by enabling them to learn more discriminative features in various emotional and acoustic conditions.

Among the most widely utilized features, **MFCCs** have been most utilized owing to their capability of representing pertinent spectral characteristics of speech signals [10]. Research like that of [11] has investigated subsets of acoustic features for machine learning-based SER, where the effectiveness of MFCCs fused with prosodic and spectral features has been examined.

Recent research has also explored new feature extraction techniques to enhance classification performance. For example, the research of [12] proposed a hybrid feature fusion technique to enhance emotion recognition by combining spectral and prosodic features, with better performance than conventional MFCC-based systems.

Early SER systems were based on classical machine learning techniques like **SVMs** and **Random Forests**. Although these techniques were effective in controlled conditions, they would tend to fail with the speech data variability caused by speaker variability and background noise.

Deep learning has significantly improved SER with the help of neural networks, especially **CNNs** and **LSTMs**. CNN-based frameworks have been efficient in feature extraction and classification, showing prowess in emotion recognition [13]. For improving performance further, hybrid models such as the **Attention-Guided 3D-CNN-LSTM** have been investigated, extracting spatiotemporal relations in speech signals [14].

More recently, Transformer-based architectures have emerged as state-of-the-art solutions in speech processing. Models like **Wav2Vec 2.0** leverage self-supervised learning and attention mechanisms to achieve superior results in SER, demonstrating remarkable improvements in feature representation and generalization [15].

2.3 Challenges and Open Issues

Despite significant advances in SER, there are some issues in building strong and generalizable systems [16][17]. These include some of the most pertinent ones:

- **Cross-Corpus Generalization:** Models trained from one set do not typically generalize to another set because there are dataset-specific biases, e.g., speaker, language, or recording conditions differences. This constraint limits the application domain of SER systems.
- **Data Imbalance:** Emotion datasets happen to be unbalanced with the majority emotion being the neutral emotion. This causes models to be biased towards the majority emotion, lowering their detection power for less frequent or more intense emotions.
- **Noise Robustness:** The systems for SER need to be robust to recording condition change as well as background noise in order to be acceptable for use in the real-world environment. Current models are not robust to noise, and that restricts their application for call center or outdoor environments.

Overcoming these issues is at the heart of making the systems more **robust and generalizable** to real-world issues.

3 Materials and Methods

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In this chapter, we show the different datasets utilized in this work, both **Italian and non-Italian corpora**, to measure the model’s ability to generalize across different languages and acoustic conditions. The major training set is **Emozionalmente**, an Italian emotional speech corpus that is released under an open source policy. To cross-check the quality of a model trained on an Italian corpus, we measure the model’s performance on **EMOVO**, another Italian corpus with different properties.

In continuing to stress the model’s resilience even further, other cross-ling datasets, namely, the German **EmoDB**, the English **RAVDESS** and **CREMA-D**, and the French **CaFE** are included as well. These cross-lingual tests provide more information on the model’s power to generalize and work across a very wide range of linguistic and acoustic conditions.

In addition, we introduce **Multilinguis**, a cross-corpus validation benchmark with heterogeneous corpora across various different languages with all the actors involved to provide a more holistic evaluation for SER models.

The chapter also covers techniques for data selection and data balancing to permit equitable and representative learning. Preprocessing operations are ultimately provided to standardize the datasets for the model in order to learn optimally.

3.1 Italian datasets

3.1.1 Emozionalmente

Emozionalmente corpus is an open-source Italian speech emotional corpus that has been obtained through crowdsourcing methods [18]. Developed with the aim to assist the SER research by providing a quality and rich corpus of emotional speech samples in the Italian language,

Table 3.1: Emozionalmente Composition and Structure

Attribute	Description
Total Samples	6,902 audio recordings
Number of Speakers	431 amateur actors
Number of Emotions	7 categories (six basic emotions + neutrality)
Emotions	Anger, Disgust, Fear, Joy, Sadness, Surprise, Neutrality
Sentences per Speaker	18 different sentences
Sampling Rate	44.1 kHz

The recordings include *acted emotional expressions*, i.e., the participants intentionally portrayed a target emotion when reading pre-defined sentences. Compared with spontaneous emotional datasets, there are several benefits in this procedure for having **clear and well-defined emotion labels**, which are suitable for supervised learning model training. The emotion labels are derived from the *intended emotion* expressed by speakers rather than perceptually validated labels from listeners.

A disadvantage of acted datasets is that they can miss **subtle emotional variations** of spontaneous speech. However, this method provides a **high degree of control over the dataset**, with balanced emotional classes and fewer noise in training data.

Emozionalmente was chosen for this research because:

1. **Comprehensive Italian SER Dataset:** One of the most extensive datasets in Italian, featuring a large number of speakers and samples.
2. **Controlled Environment:** Ensures well-defined labels and minimal ambiguity, which is critical for training emotion recognition models.
3. **High-Quality Audio:** Provides recordings with minimal background noise, improving feature extraction reliability.

3.1.2 EMOVO

The *EMOVO* dataset is a widely used Italian emotional speech dataset, devised particularly to advance research in SER [19]. The dataset provides a benchmark by which models can be evaluated on Italian emotional speech and provides a controlled setting in which to study various classes of emotions. The dataset has been recorded in a **controlled studio** so that the recording would be of good quality and the recording artifacts and the background noise would be as minimal as possible.

Table 3.2: EMOVO Composition and Structure

Attribute	Description
Total Samples	588 audio recordings
Number of Speakers	6 native Italian speakers (3 male, 3 female)
Number of Emotions	7 categories (six basic emotions + neutrality)
Emotions	Anger, Disgust, Fear, Joy, Sadness, Surprise, Neutrality
Sentences per Speaker	7 different sentences
Sampling Rate	48 kHz

The EMOVO database has been recorded with professional recording equipment in an anechoic room with the view towards **acoustic purity**. The same sentences, with the same number of emotions per speaker, were recorded by all six speakers.

One distinctive feature of the resulting dataset is the **emotion annotation process**. Unlike other databases relying on speaker-tagged emotion only, EMOVO integrated the verification process wherein human annotators checked if the perceived emotion and intended label match. This makes the dataset more credible when used for training and testing SER systems.

The EMOVO dataset was selected for the present study because of the following major features:

1. **Standard Benchmark:** Widely used in Italian SER research, making it an ideal test set for evaluating model performance.
2. **High-Quality Studio Recordings:** Provides clean and noise-free speech samples, reducing the impact of external factors.
3. **Small but Well-Controlled Dataset:** The limited number of speakers ensures uniformity and consistency in emotional expressions.

3.2 English datasets

3.2.1 RAVDESS

The *Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS)* is an openly available, English language SER dataset [20]. The database has high-quality audio records of emotion-expressing speech spoken and sung and therefore, can be utilized for emotion classification.

Table 3.3: RAVDESS Composition and Structure

Attribute	Description
Total Samples	1,440 audio recordings
Number of Speakers	24 (12 male, 12 female)
Number of Emotions	8 categories (7 emotions + neutrality)
Emotions	Anger, Disgust, Fear, Joy, Sadness, Surprise, Calm, Neutrality
Expression Modes	Speech and Song
Emotion Intensity	2 levels (Normal, Strong)
Sentence Variability	24 different sentences
Sampling Rate	48 kHz

The database has *acted emotional expressions* by professional actors. Every speaker speaks two different sentences with varying degrees of emotion (*normal* and *strong*). A plus with RAVDESS is that it has good, well-tagged samples with explicit emotional expressions. A minus, however, is that because the database has acted emotions, it may not be capable of capturing the spontaneous emotional dynamics involved in natural speech. Main points:

1. **Provide a Benchmark for English SER:** As a widely used dataset, it facilitates comparisons with prior research.
2. **Support Cross-Language Evaluation:** Enables the assessment of the Italian-trained model on English speech.
3. **Ensure High-Quality Emotional Expressions:** The dataset's professional recordings help mitigate noise-related issues.

3.2.2 CREMA-D

The *Crowd-Sourced Emotional Multimodal Actors Dataset*, or **CREMA-D**, refers to an English emotion corpus that has been created to enable research on SER [21]. The dataset has many different speakers and expressions that convey different emotions.

Table 3.4: CREMA-D Composition and Structure

Attribute	Description
Total Samples	7,442 audio recordings
Number of Speakers	91 actors (48 male, 43 female)
Number of Emotions	6 categories (5 emotions + neutrality)
Emotions	Anger, Disgust, Fear, Joy, Sadness, Neutrality
Sentence Variability	12 different sentences
Sampling Rate	44.1 kHz

In contrast to the majority of SER datasets, CREMA-D was annotated with *crowd-sourced perceptual validation* where human annotators indicated the emotional expressions from the perceived emotions and not intended expressions. This renders the dataset more credible for real-world application.

One of the key strengths of the CREMA-D set is that it possesses a **large speaker base** which helps to generalize the model across voices. Nevertheless, since the set is created from a limited number of written words, it might not have the variability of spontaneous emotional speech.

CREMA-D is included in this study to:

1. **Assess Generalization Across Diverse Speakers:** The high number of speakers aids in evaluating speaker variability.
2. **Support Cross-Language Experiments:** Provides additional insights into how the model performs on non-Italian datasets.
3. **Validate Emotion Recognition in Perceptually Annotated Data:** Ensures the model is trained and evaluated on emotions as perceived by listeners.

3.3 German datasets

3.3.1 EmoDB

The *Berlin Database of Emotional Speech (EmoDB)* is a German corpus that has found widespread application in SER research [22]. Professional actors read emotional speech samples, and it gives valuable samples for emotion recognition system training and testing.

Table 3.5: EmoDB Composition and Structure

Attribute	Description
Total Samples	535 audio recordings
Number of Speakers	10 actors (5 male, 5 female)
Number of Emotions	7 categories (6 emotions + neutrality)
Emotions	Anger, Boredom, Fear, Joy, Sadness, Disgust, Neutrality
Sentence Variability	10 different sentences
Sampling Rate	16 kHz

In contrast to most of the SER databases, EmoDB is founded on *acted emotional speech*, wherein ten professional actors, five male and five female, read aloud ten German sentences of varying emotions. The recordings took place in an anechoic chamber at the Technical University Berlin, offering the high-quality recording quality.

One of the strongest points about EmoDB is its **clear emotional labeling** on each utterance with the target emotion. However, as the corpus uses acted speech, it may not reflect the range of emotional expressions that happen spontaneously.

EmoDB is included in this study to:

1. **Evaluate Emotion Recognition in German Speech:** Provides a benchmark for testing models on German emotional speech.
2. **Analyze Acted Emotional Speech:** Allows examination of models trained on controlled, acted emotional expressions.
3. **Compare Performance Across Datasets:** Facilitates assessment of model generalization by comparing results with other emotion datasets.

3.4 French datasets

3.4.1 CaFE

The *Canadian French Emotional Speech Dataset* (CaFE) is a dataset for the French language with the purpose to make SER easier [23]. The dataset has emotional utterances that are acted by professional actors, providing valuable assets for emotion recognition system training and testing.

Table 3.6: CaFE Composition and Structure

Attribute	Description
Total Samples	936 audio recordings
Number of Speakers	12 actors (6 male, 6 female)
Number of Emotions	7 categories (6 basic emotions + neutrality)
Emotions	Anger, Disgust, Fear, Joy, Sadness, Surprise, Neutrality
Sentence Variability	6 different sentences
Emotion Intensity	2 levels (Normal, Strong)
Sampling Rate	48/192 kHz

CaFE corpus uses *acted emotional speech* and has been read by 12 professional actors, having 6 sentences each, respectively, with two intensities per emotion: normal and strong. They have been recorded using a sample frequency of 192 kHz (24 bits per sample), and subsequently filtered and downsampled with a lower resolution using a sample frequency of 48 kHz, with samples using 16 bits per sample.

One of the major strengths of CaFE is that it has **clear emotional labeling**, as each utterance is labeled with the target emotion. Since the corpus, though, is made up of acted speech, it may not reflect the variation of spontaneous emotional speech.

CaFE is included in this study to:

1. **Evaluate Emotion Recognition in French Speech:** Provides a benchmark for testing models on French emotional speech.
2. **Analyze Acted Emotional Speech:** Allows the examination of models trained on controlled, acted emotional expressions.
3. **Compare Performance Across Datasets:** Facilitates assessment of model generalization by comparing results with other emotion datasets.

3.5 Multilingual Dataset

3.5.1 Multilinguis

Since standardized multilingual datasets for SER do not yet exist, and the emotional tags available in existing datasets lack uniformity, we formed **Multilinguis** to benchmark cross-corpus validation. Multilinguis consolidates emotion-tagged audio across five datasets (RAVDESS, EMOVO, EmoDB, CREMA-D and CaFe) into a heterogeneous, multilingual corpus to evaluate SER models in cross-corpus as well as cross-lingual conditions.

Table 3.7: Multilingual SER Dataset (Multilinguis) Composition and Structure

Attribute	Description
Total Samples	2,987 audio recordings
Number of Languages	4 languages: Italian, English, German, French
Source Datasets	5 (EMOVO, RAVDESS, CREMA-D, EmoDB, CaFe)
Number of Speakers	143 actors (74 male, 69 female)
Number of Emotions	7 categories (six basic emotions + neutrality)
Emotions	Anger, Disgust, Fear, Joy, Sadness, Surprise, Neutrality
Sentence Variability	59 different sentences
Sampling Rate	16 kHz to 48 kHz

The dataset includes up to 100 audio samples per emotion per corpus, covering all actors from the source datasets. By selecting samples across multiple languages, it ensures high-quality, emotion-labeled data for SER tasks. This approach balance linguistic diversity and speaker consistency across datasets.

The dataset is ideal for the following purposes:

1. **Benchmarking SER Models Across Multiple Languages:** By including data from RAVDESS, EMOVO, EmoDB, CREMA-D, and CaFe, the dataset provides a robust platform for testing models on cross-corpus emotion recognition tasks.
2. **Cross-Corpus Validation:** The dataset is designed to evaluate the generalization ability of SER models when applied to different speech corpora.
3. **Support for Cross-Lingual and Multi-Lingual Emotion Recognition:** The diversity in language and emotion ensures that the dataset can be used for testing models in real-world, multilingual scenarios.

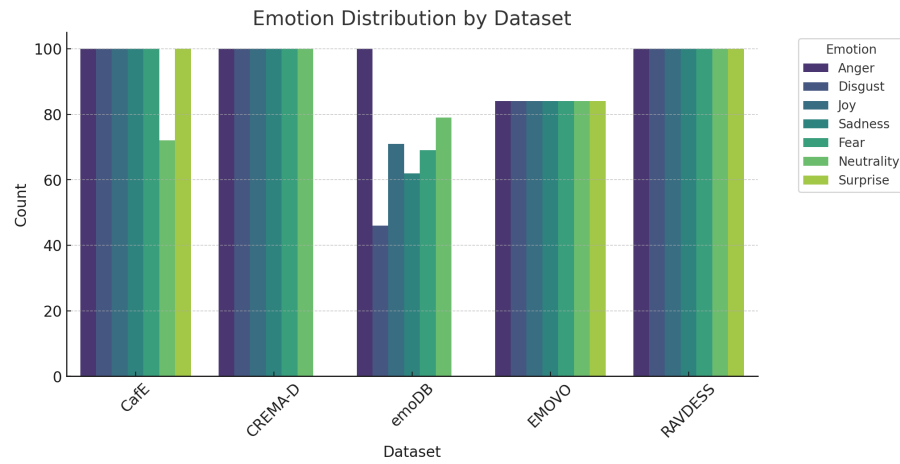


Figure 3.1: Emotion Distribution by Dataset

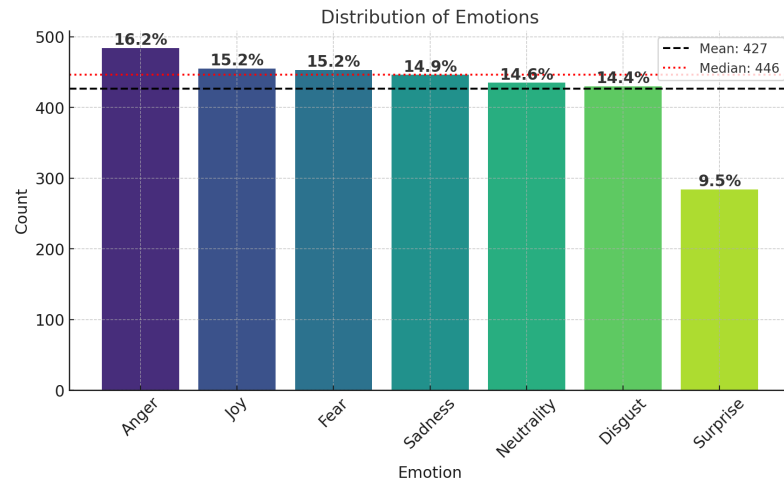


Figure 3.2: Distribution of Emotions

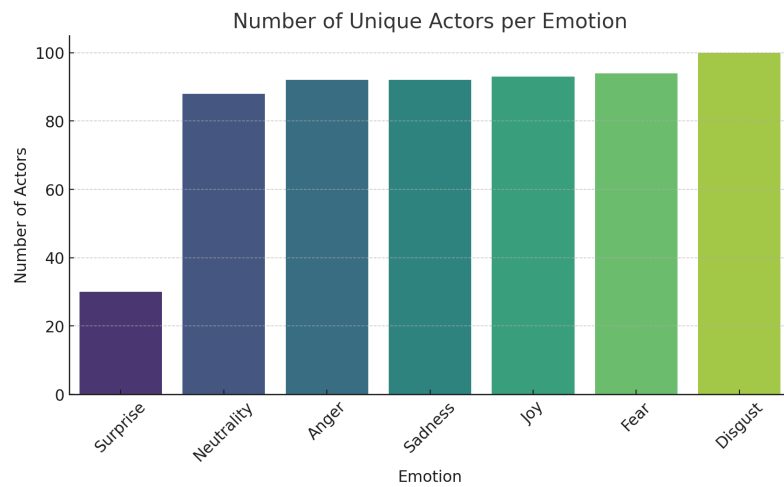


Figure 3.3: Number of Unique Actors per Emotion

3.6 Data Selection and Balancing Techniques

For this study, we selected the *Emozionalmente* dataset as the primary set due to the quality of the emotional annotation, with the ratings performed by five different annotators per each audio sample.

Previous research has shown that humans identify emotions in speech with an accuracy of 65% [24]. In the *Emozionalmente* dataset, it is verified the same result, with the annotated-perceived emotion congruence being 66%. Training a model on weak labels, where annotations are influenced by human perception, inherently transfers the bias of human emotion recognition capabilities to the model itself. Based on this observation, we obtained a hard core subset of the dataset by iteratively selecting files with the maximum annotation congruence.

The data selection had been planned in three rounds:

- **Round 1:** Selected only the files where all five raters unanimously agreed on the expressed emotion, ensuring maximum annotation consistency.
- **Round 2:** Included additional files where at least four out of five raters agreed, allowing for a slightly broader but still highly reliable subset.
- **Round 3:** Further expanded the dataset by incorporating files where at least three out of five raters agreed, balancing dataset size with annotation consistency.

Table 3.8: Number of selected audio files per emotion and annotation round.

Emotion	Round 1	Round 2	Round 3
Disgust	90	165	253
Joy	228	237	43
Fear	116	188	204
Anger	260	248	0
Surprise	292	216	0
Sadness	289	219	0
Neutral	508	0	0

After selection, we balanced the datasets by equaling the number of samples per emotion, ending up with a final 508 samples per class and a total of 3556 audio files.

Also, we utilised several data augmentation techniques to improve the model robustness and generalizability. The augmentation techniques that we used to mimic real-world distortions without destroying emotional cues included the following:

- **Time Stretching:** Varying the playback speed within a range of 10%.
- **Gaussian Noise Injection:** Adding small random noise to simulate background disturbances.
- **Vocal Tract Length Perturbation (VTLP):** Altering the spectral characteristics to simulate variations in speakers' vocal tract.
- **Room Impulse Response (RIR) Augmentation:** Applying real reverberation effects to improve model generalization in noisy environments.

We utilized real recorded impulse responses for RIR-based augmentation because the literature has shown their ability to enhance real-world cross-corpus generalization [25].

After applying data augmentation, the kernel extracted from the Emozionalmente dataset was pushed to a total of **17,780 audio samples**. The dataset size was augmented by a factor of 5 as a consequence. Data augmentation techniques provided an evenly distributed and varied dataset, minimizing the risk of overfitting and enhancing the model's generalizability to real-world use.

3.7 Data Preparation and Preprocessing

As this project is directed towards assessing the feasibility of an SER model in practical application, we need to train the model in a **speaker-independent** manner. For this purpose, we use `StratifiedGroupKFold`, which provides balanced and speaker-independent data splitting. This approach avoids **data leakage** as it does not permit the same speaker in training, validation, and test datasets. Additionally, it preserves class balance between folds, thereby limiting the scope for imbalance-based biases.

With the speaker-independent split, the model is made to generalize better to new speakers by not overfitting to speaker-dependent traits. Various random states are attempted to achieve the best class distribution, choosing the optimal split by reducing imbalance. The process line-by-line is presented in the given pseudocode:

Algorithm 1: Best Balanced Data Split Algorithm

Input: Dataset df , maximum attempts $max_attempts$, seed $seed$

Output: Best balanced split ($train_df, val_df, test_df$)

Initialize $best_balance \leftarrow \infty$, $best_splits \leftarrow None$;

Set random seed using $seed$;

Generate $max_attempts$ random states;

foreach $random_state$ in $random_states$ **do**

 Perform first stratified group split (5-fold);

 Select first split: $train_idx, temp_idx$;

 Extract validation-test subset val_test_data from $temp_idx$;

 Perform second stratified group split (2-fold) on val_test_data ;

 Select first split: $val_idx, test_idx$;

 Create $train_df, val_df, test_df$ using indices;

 Compute balance score as the sum of standard deviations:

$$balance_score = \sigma(train_df) + \sigma(val_df) + \sigma(test_df)$$

 where $\sigma(X)$ is the standard deviation of class distribution in X ;

if $balance_score < best_balance$ **then**

 | Update $best_balance$ and $best_splits$;

end

end

return $best_splits$;

Figure 3.4: Best Balanced Data Split Algorithm

The following tables summarize the dataset sizes and class distributions after the best split was selected.

Set	Number of Samples	Percentage of Total (%)	Imbalance Ratio
Train	14670	82.51%	1.02
Validation	1401	7.87%	1.17
Test	1709	9.61%	1.16

Table 3.9: Dataset sizes, relative percentages, and imbalance ratios for each split.

Class	Train (%)	Validation (%)	Test (%)
Anger	14.26	14.13	14.63
Disgust	14.25	13.85	14.92
Fear	14.18	14.99	14.63
Joy	14.4	14.92	12.81
Neutral	14.38	12.78	14.69

Table 3.10: Class distribution across train, validation, and test sets.

The previous tables illustrate that the dataset split realizes a **speaker-independent and balanced** split, most importantly for the sake of stable model performance. The class distributions in the training, validation, and test sets are balanced, which reduces the risk of class imbalance and the validity of the performance estimation to a minimum. In addition, ratios of imbalance close to 1 imply that no class outweighs the others, thus reducing bias and improving the model’s capacity to generalize. This carefully crafted data split implies that our SER model is trained on diverse and representative samples, which makes it ultimately more robust in real-world scenarios.

4 Model Architecture

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4.1 Fine-tuning of Wav2Vec2-Large-XLSR-53-Italian	23
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Latest developments in SER leverage utilize **Transformer-based models** which surpass conventional CNNs and RNNs by modeling long-range dependencies and contextual info in speech signals. They enable them to be especially effective at extracting emotional info from raw audio directly.

4.1 Fine-tuning of Wav2Vec2-Large-XLSR-53-Italian

Our model is founded on **Wav2Vec2 2.0 large**, a pre-trained self-supervised speech representation learning model that takes raw waveforms as input without expertly crafted features such as MFCCs [26]. Wav2Vec2 learns powerful representations using contrastive loss and a Transformer encoder, facilitating strong generalization to numerous datasets and languages.

In order to replicate the experiment previously conducted by dr. Fabio Catania with the *Emozionalmente* dataset [18] and to perform a direct comparison to evaluate whether the techniques used lead to improved performance, we fine-tuned the jonatasgrosmann/wav2vec2-large-xlsr-53-italian model [27], a multilingual variant of Wav2Vec2 pre-trained on **53 languages** as part of the XLSR (Cross-Lingual Speech Representations) initiative. This model was further fine-tuned on the **Italian subset of Mozilla Common Voice 6.1** for Automatic Speech Recognition (ASR), making it particularly well-suited for speech tasks in Italian.

The fine-tuning is done by training the model on our train dataset and freezing the lower convolutional layers of Wav2Vec2 in order to maintain the pre-learned speech representations. We add a fully connected classification head and fine-tune the whole model with a **cross-entropy loss function** with the AdamW optimizer.

The training goes in the following steps:

1. **Audio Preprocessing:** Samples are resampled to 16 kHz and trimmed or zero-padded to 4 seconds to have a fixed input length.
2. **Feature Extraction:** Wav2Vec2's processor converts raw waveforms into contextual embeddings.
3. **Classification Head:** A fully connected layer with **7 output neurons** maps Transformer outputs to logits, which are converted to class probabilities using a **softmax function** during inference.
4. **Optimization:** The model is trained using the **AdamW** optimizer and a **learning rate of 5×10^{-5}** , batch size **32**, and **10 epochs**. Weight decay is **0.01** for regularization of the model.
5. **Training Strategy:** The model is trained with an **evaluation strategy per epoch**, saving the best model based on `eval_loss`. Mixed precision training is also enabled with **FP16** for greater efficiency, and **gradient checkpointing** is used for memory conservation. Data loading is accelerated with **2 worker threads**. The best model is loaded automatically at the conclusion of training.

Epoch	Training Loss	Validation Loss	Accuracy
1	0.515600	0.414341	0.865810
2	0.207600	0.421463	0.879372
3	0.256300	0.351275	0.907209
4	0.066400	0.346235	0.919343
5	0.023900	0.492293	0.898644
6	0.029900	0.456569	0.908637
7	0.030400	0.295209	0.941470
8	0.039200	0.436384	0.932191
9	0.060100	0.333166	0.942898
10	0.029200	0.333144	0.942898

Table 4.1: Training and Validation Results

5 Results

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In this chapter, we present a full evaluation of our fine-tuned model on a variety of datasets with various linguistic and acoustic properties. We try to verify the model’s robustness and whether or not it can preserve high performance across various recording conditions, speaker populations, and emotional distributions.

We begin by announcing the primary test set used for testing and then move to further the assessment to popular SER datasets such as EMOVO (Italian), RAVDESS (English), CREMA-D (English), EmoDB (German), and CaFE (French). We also introduce **Multilinguis**, a multilingual benchmark dataset created for this study. By incorporating a diverse set of corpora, we aim to provide a comprehensive study of the strengths and weaknesses of the model, particularly in cross-corpus scenarios where the performance would otherwise degrade.

The outcomes of these tests will dictate the flexibility of the model and what can be further improved on to increase its usefulness in practical reality.

5.1 Test Set

Our analysis shows that the model achieves **93.56% accuracy** in SER, with **93.5% UA/WA** and an MCC of **0.9251**. The applied techniques improved performance by 13% over the previous study [18]. The confusion matrix reveals minimal misclassification, with per-emotion accuracy ranging from **89.02% to 99.58%**. A **0.9329** correlation with ground truth further confirms its reliability.

Metric	Value
Accuracy	93.56%
Precision (Macro Avg)	94.00%
Recall (Macro Avg)	93.54%
F1-Score (Macro Avg)	93.53%
Weighted F1-Score	93.56%
Unweighted Average Recall (UA)	93.54%
Weighted Average Recall (WA)	93.56%
Matthews Correlation Coefficient (MCC)	0.9251
Ground Truth - Prediction Correlation	0.9329

Emotion	Accuracy
Anger	94.80%
Disgust	91.37%
Joy	89.95%
Sadness	89.02%
Fear	93.60%
Surprise	99.58%
Neutral	96.41%

Table 5.1: Test Set Performance Metrics

Table 5.2: Emotion Accuracy on Test Set

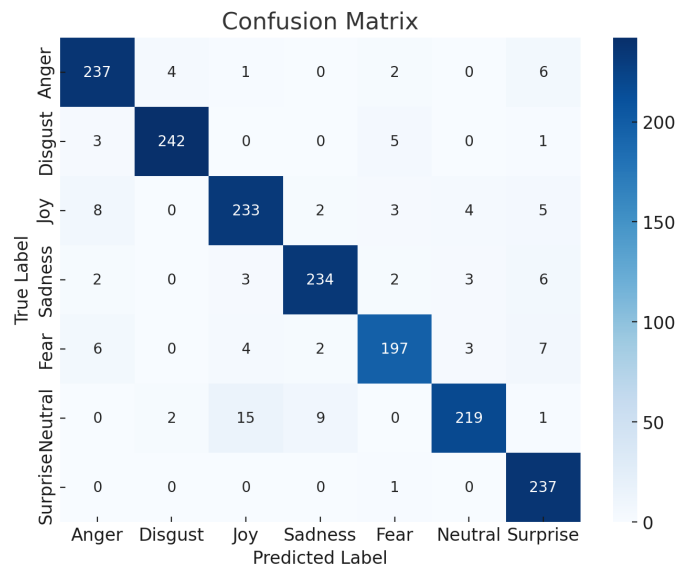


Figure 5.1: Confusion Matrix of the Test Set

5.2 EMOVO

To assess generalization, we tested our model on **EMOVO**. Accuracy reached **61.73%**, highlighting moderate domain adaptation. Although **Anger** remained high (91.67%), emotions like **Sadness** (48.81%), **Disgust** (44.05%) and **Joy** (46.43%) saw declines, suggesting differences in expression or recording conditions.

Metric	Value
Accuracy	61.73%
Precision (Macro Avg)	66.13%
Recall (Macro Avg)	61.73%
F1-Score (Macro Avg)	61.12%
Weighted F1-Score	61.02%
Unweighted Average Recall (UA)	61.73%
Weighted Average Recall (WA)	61.73%
Matthews Correlation Coefficient (MCC)	0.5639
Ground Truth - Prediction Correlation	0.5677

Emotion	Accuracy
Anger	91.67%
Disgust	44.05%
Joy	46.43%
Sadness	48.81%
Fear	63.10%
Surprise	52.38%
Neutral	85.71%

Table 5.3: EMOVO Performance Metrics

Table 5.4: Emotion Accuracy on EMOVO

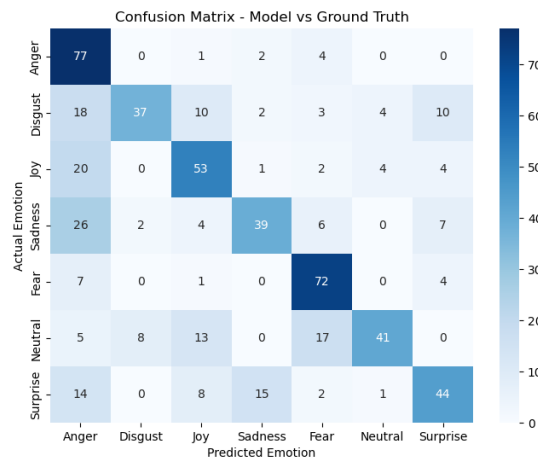


Figure 5.2: Confusion Matrix of EMOVO

5.3 RAVDESS

To assess generalization, we tested our model on **RAVDESS**. Accuracy reached **60.27%**, showing moderate domain adaptation. While **Surprise** (81.25%) and **Anger** (80.21%) maintained high accuracy, emotions like **Neutral** (38.54%) and **Disgust** (40.10%) exhibited a notable drop, suggesting variations in emotional expression or dataset differences.

Metric	Value
Accuracy	60.27%
Precision (Macro Avg)	60.30%
Recall (Macro Avg)	60.27%
F1-Score (Macro Avg)	59.31%
Weighted F1-Score	59.04%
Unweighted Average Recall (UA)	60.27%
Weighted Average Recall (WA)	60.27%
Matthews Correlation Coefficient (MCC)	0.5396
Ground Truth - Prediction Correlation	0.5951

Table 5.5: RAVDESS Performance Metrics

Emotion	Accuracy
Anger	80.21%
Disgust	40.10%
Joy	60.42%
Sadness	56.77%
Fear	64.58%
Surprise	81.25%
Neutral	38.54%

Table 5.6: Emotion Accuracy on RAVDESS

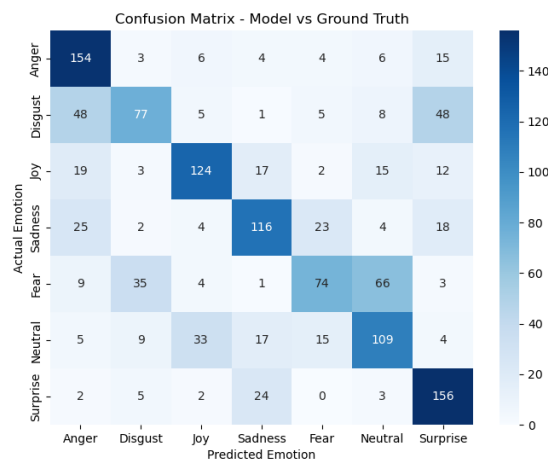


Figure 5.3: Confusion Matrix of RAVDESS

5.4 CREMA-D

To assess generalization, we tested our model on **CREMA-D**. Accuracy reached **57.20%**, indicating moderate domain adaptation. While **Neutral** (76.50%) and **Disgust** (64.29%) maintained high accuracy, emotions like **Anger** (51.70%) and **Fear** (33.00%) showed notable variations, suggesting dataset differences or variations in emotional expression.

Metric	Value
Accuracy	57.20%
Precision (Macro Avg)	58.00%
Recall (Macro Avg)	57.34%
F1-Score (Macro Avg)	56.50%
Weighted F1-Score	56.03%
Unweighted Average Recall (UA)	57.34%
Weighted Average Recall (WA)	57.20%
Matthews Correlation Coefficient (MCC)	0.4896
Ground Truth - Prediction Correlation	0.5505

Emotion	Accuracy
Anger	51.70%
Disgust	64.29%
Joy	63.80%
Sadness	54.76%
Fear	33.00%
Neutral	76.50%

Table 5.7: CREMA-D Performance Metrics

Table 5.8: Emotion Accuracy on CREMA-D

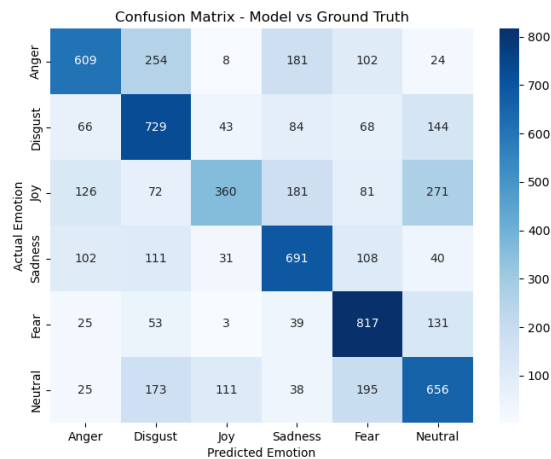


Figure 5.4: Confusion Matrix of CREMA-D

5.5 EmoDB

To evaluate the model's generalization to German-language speech, we tested it on **EmoDB**. The model achieved an **accuracy of 76.71%**, demonstrating strong domain adaptation. Notably, **Fear (95.38%)** and **Joy (90.00%)** were classified with high accuracy, while **Anger (66.39%)** and **Neutral (66.67%)** showed lower performance, suggesting differences in emotional expressiveness or dataset characteristics.

Metric	Value
Accuracy	76.71%
Precision (Macro Avg)	81.00%
Recall (Macro Avg)	77.99%
F1-Score (Macro Avg)	77.22%
Weighted F1-Score	77.00%
Unweighted Average Recall (UA)	77.99%
Weighted Average Recall (WA)	76.71%
Matthews Correlation Coefficient (MCC)	0.7285
Ground Truth - Prediction Correlation	0.7933

Table 5.9: EmoDB Performance Metrics

Emotion	Accuracy
Anger	66.39%
Disgust	70.45%
Joy	90.00%
Sadness	79.03%
Fear	95.38%
Neutral	66.67%

Table 5.10: Emotion Accuracy on EmoDB

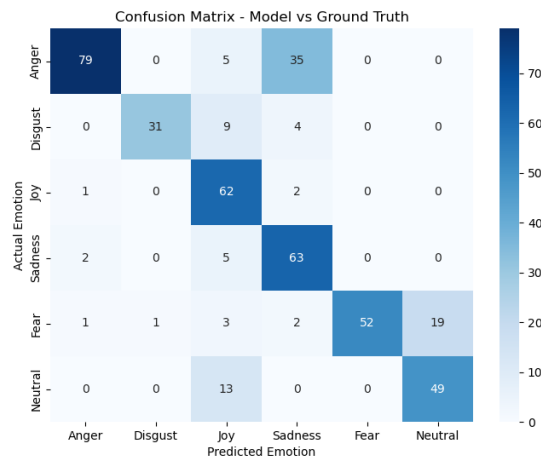


Figure 5.5: Confusion Matrix of EmoDB

5.6 CaFE

To assess generalization, we tested our model on **CaFE**. Accuracy reached **64.96%**, indicating moderate domain adaptation. While **Surprise** (86.81%) and **Fear** (81.94%) maintained high accuracy, emotions like **Joy** (31.25%) and **Disgust** (62.50%) showed notable variations, suggesting dataset differences or variations in emotional expression.

Metric	Value
Accuracy	64.96%
Precision (Macro Avg)	69.12%
Recall (Macro Avg)	64.38%
F1-Score (Macro Avg)	64.80%
Weighted F1-Score	64.12%
Unweighted Average Recall (UA)	64.38%
Weighted Average Recall (WA)	64.96%
Matthews Correlation Coefficient (MCC)	0.5941
Ground Truth - Prediction Correlation	0.6939

Emotion	Accuracy
Anger	66.67%
Disgust	62.50%
Joy	31.25%
Sadness	64.58%
Fear	81.94%
Neutral	56.94%
Surprise	86.81%

Table 5.11: CaFE Performance Metrics

Table 5.12: Emotion Accuracy on CaFE

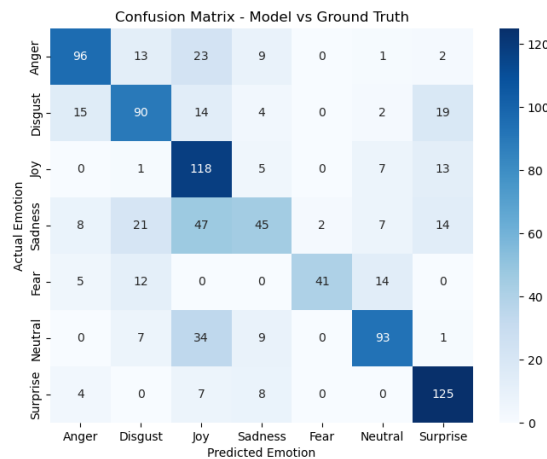


Figure 5.6: Confusion Matrix of CaFE

5.7 Multilinguis

To assess generalization across languages, we tested our model on Multilinguis. Accuracy reached **61.53%**, indicating moderate domain adaptation. While **Surprise** (74.30%) and **Anger** (68.80%) maintained high accuracy, emotions like **Joy** (54.73%) and **Disgust** (52.09%) showed notable variations, suggesting dataset differences or variations in emotional expression.

Metric	Value
Accuracy	61.53%
Precision (Macro Avg)	62.11%
Recall (Macro Avg)	62.09%
F1-Score (Macro Avg)	61.45%
Weighted F1-Score	61.23%
Unweighted Average Recall (UA)	62.09%
Weighted Average Recall (WA)	61.53%
Matthews Correlation Coefficient (MCC)	0.5516
Ground Truth - Prediction Correlation	0.5795

Table 5.13: Multilinguis Performance Metrics

Emotion	Accuracy
Anger	68.80%
Disgust	52.09%
Joy	54.73%
Sadness	58.07%
Fear	62.25%
Neutral	64.37%
Surprise	74.30%

Table 5.14: Emotion Accuracy on Multilinguis

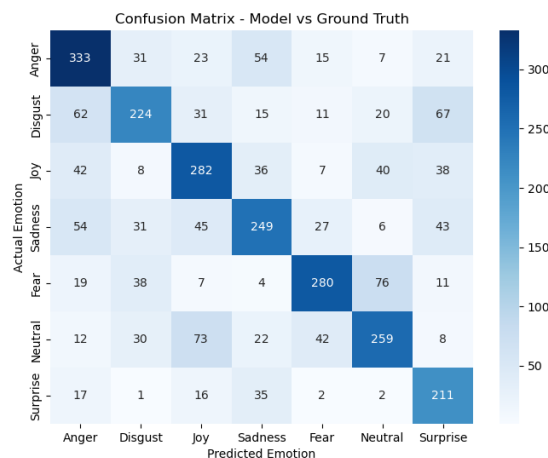


Figure 5.7: Confusion Matrix of Multilinguis

6 emoBOX Benchmark

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6.1 emoBOX

SER is crucial for human-computer interaction but hindered by inconsistent dataset splits and lack of benchmarks. To address this, **emoBOX** provides a multilingual toolkit and a large benchmark for evaluating SER models in intra and cross-corpus settings [28].

EmoBOX also supports pre-computed dataset splits for intra-corpus experiments and new test set balancing for cross-corpus setups. It is provided with *emotion2vec* for balancing annotation disparities for equitable comparisons. EmoBOX, consisting of 32 emotion datasets across 14 languages, is the largest SER benchmark, enabling reproducibility and research.

To validate model generalization, the study attempted 10 Transformer-based models under cross-corpus validation settings. The models were evaluated on different datasets to examine their robustness and generalizability to new emotional speech distributions not seen before. The outcome of this test is presented in the following table:

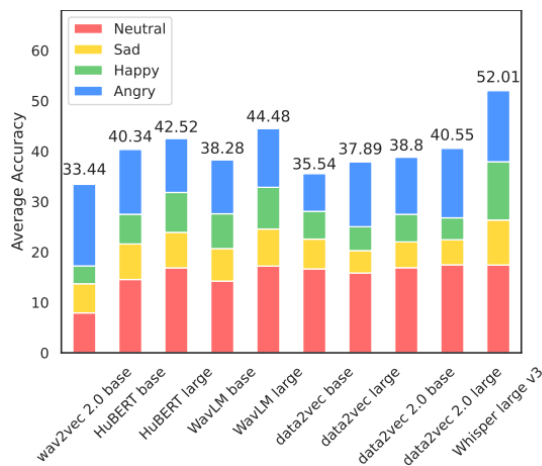


Figure 6.1: Overview of the emoBOX benchmark.

The best-performing Transformer model in the benchmark is **Whisper large V3**, with an average accuracy of **52.01%**, which outperforms the other models in the experiment by a large margin. To compare, **wav2vec 2.0 base**, one of the baseline models, reaches an average accuracy of only **33.44%**, with a large performance difference from Whisper large V3.

The better performance of **Whisper large V3** is due to its large-scale pretraining on diverse speech corpora, which allows it to generalize better to other corpora [29]. Its novel architecture also incorporates better speech representation learning, allowing it to be more resilient to recording quality variations, speaker variation, and emotional intensity.

On the other hand, benchmarks on **wav2vec 2.0 base** and **data2vec** show lower accuracy because of the decreased training range and reduced flexibility in conforming to cross-corpus variability. These models can perform some speech recognition tasks but are behind in emotion recognition when the datasets considered are considerably different from their training distributions.

6.2 Interpreting the Results

In our analysis, we have chosen a sub-group of **10 datasets** from the original **32-dataset** emoBOX benchmark because of **access limitations** on a few corpora and the **unavailability of obtaining certain datasets**. The selection enables us to have a fair model performance comparison with a varied dataset selection.

However, sampling only a subset creates a **margin of error**, which is the amount of potential variation of our estimate from the entire benchmark. Margin of error (e) for a finite population is derived by:

$$e = Z \times \sqrt{\frac{p(1-p)}{n} \times \frac{N-n}{N-1}} \quad (6.1)$$

where:

- $N = 32$ is the total number of datasets in the original benchmark,
- $n = 10$ is the number of datasets used in our evaluation,
- $Z = 1.96$ corresponds to a 95% confidence level,
- $p = 0.5$ represents the maximum variability assumption.

Using this formula, the margin of error in our study is **26.50%**. So, although our results have important conclusions about model performance, the results must be taken with some caution since running the complete benchmark could result in somewhat different rankings.

To mitigate this limitation, we are interested in relative model performance trends instead of absolute accuracy. Relative difference in performance attained within our chosen subset of dataset continues to be representative of overall trend and with given dataset diversity preserved.

6.3 Results

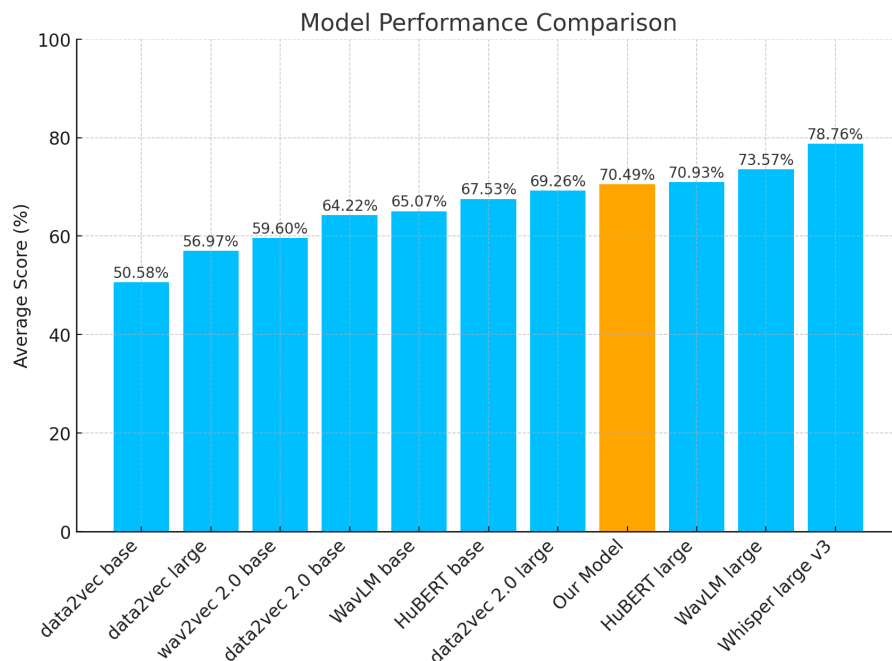


Figure 6.2: Average Accuracy per Model on the EmoBox Benchmark.

From the comparison, we observe that:

- Our model (**wav2vec 2.0 large**) achieves a significant improvement over its base counterpart (+**10.89%** accuracy) due to its increased depth (**24 layers**) and parameter count (**317M**).
- **Data2Vec 2.0 base** and **HuBERT base** perform better than wav2vec 2.0 base, likely due to their distinct self-supervised training strategies.
- The **large models** (ours, Data2Vec 2.0 large, HuBERT large and WavLM large) benefit from deeper architectures, with WavLM large achieving the highest accuracy (**73.57%**) due to speaker-aware pretraining.
- **Whisper large V3** surpasses all models in accuracy (**78.76%**), but its **1.55B parameters** make it significantly more resource-intensive, limiting its practicality for SER.

Our model balances accuracy and efficiency, performing competitively while avoiding excessive computational costs.

Table 6.1: Performance comparison of different models across multiple datasets on emoBOX benchmark.

Model	CREMA-D (en)			EMOVO (it)			RAVDESS (en)			EmoDB (de)			CaFE (fr)		
	UA (%)	WA (%)	F1 (%)	UA (%)	WA (%)	F1 (%)	UA (%)	WA (%)	F1 (%)	UA (%)	WA (%)	F1 (%)	UA (%)	WA (%)	F1 (%)
wav2vec 2.0 base	61.95	61.90	61.75	31.07	31.07	27.24	54.33	55.40	53.99	82.06	83.14	82.21	42.76	42.47	41.03
HuBERT base	71.13	70.98	71.00	46.10	46.10	41.28	65.43	66.21	65.31	87.73	87.73	87.82	54.16	54.16	53.36
HuBERT large	73.83	73.64	73.73	45.74	45.74	40.56	70.00	70.29	69.54	89.81	90.26	89.86	59.50	58.73	58.22
WavLM base	69.64	69.49	69.54	42.39	42.39	37.33	61.56	62.10	61.18	87.03	87.12	86.76	52.71	52.33	51.66
WavLM large	74.50	74.32	74.39	48.82	48.82	44.16	72.00	72.22	71.42	92.58	92.67	92.57	62.20	61.33	61.14
data2vec base	58.03	57.78	57.73	32.47	32.47	29.22	51.92	52.22	51.10	58.12	60.01	58.32	42.18	42.36	41.69
data2vec large	63.80	63.51	63.48	35.66	35.66	33.59	59.30	59.50	58.74	60.96	61.95	61.26	42.24	42.85	41.11
data2vec 2.0 base	65.74	65.48	65.47	42.96	42.96	41.01	64.66	64.84	64.18	75.07	75.86	75.49	51.83	51.67	50.52
data2vec 2.0 large	69.55	69.27	69.25	45.88	45.88	43.63	71.15	71.63	70.94	79.36	80.41	79.96	59.04	58.02	57.51
Whisper large v3	76.75	76.48	76.60	57.82	57.82	56.06	75.32	75.87	75.19	91.26	92.43	91.84	69.43	68.84	68.06
Our Model	72.25	73.51	72.95	63.62	67.50	65.39	74.58	76.70	74.86	86.17	85.08	86.35	77.41	77.14	76.17
Model	SUBESCO (bn)			URDU (ur)			ShEMO (fa)			ASED (am)			Oreau (fr)		
	UA (%)	WA (%)	F1 (%)	UA (%)	WA (%)	F1 (%)	UA (%)	WA (%)	F1 (%)	UA (%)	WA (%)	F1 (%)	UA (%)	WA (%)	F1 (%)
wav2vec 2.0 base	51.25	51.25	50.91	87.50	87.50	87.57	56.34	78.96	57.34	88.59	88.63	88.63	40.14	41.06	39.23
HuBERT base	57.89	57.89	57.64	88.41	88.41	88.40	58.31	81.17	63.15	94.17	94.13	94.13	51.98	52.80	51.89
HuBERT large	64.53	64.53	64.33	81.75	81.75	81.66	64.29	83.35	66.26	96.19	96.19	96.17	63.69	64.35	63.66
WavLM base	57.06	57.06	56.78	82.82	82.82	82.85	60.73	78.76	62.06	94.27	94.31	94.29	57.78	58.06	57.45
WavLM large	65.33	65.33	65.09	86.61	86.61	86.64	71.72	87.13	73.55	96.44	96.45	96.42	65.54	66.29	65.67
data2vec base	46.22	46.22	45.80	65.42	65.42	65.35	47.61	70.07	49.49	86.39	86.34	86.34	54.84	55.67	54.76
data2vec large	53.06	53.06	52.62	64.81	64.81	65.35	56.42	74.09	60.98	88.31	88.31	88.30	48.29	49.12	48.21
data2vec 2.0 base	57.97	57.97	57.69	69.97	69.97	69.91	60.59	79.03	64.03	93.99	93.99	93.98	59.40	59.73	58.13
data2vec 2.0 large	66.45	66.45	66.25	78.10	78.10	78.10	64.09	82.68	68.47	94.30	94.28	94.27	64.64	64.88	64.50
Whisper large v3	73.05	73.05	72.94	82.52	82.52	82.41	80.23	89.55	82.94	96.75	96.73	96.74	84.48	84.79	85.01
Our Model	61.42	67.46	63.15	57.22	63.36	54.81	76.08	81.05	74.99	61.86	66.64	61.65	74.29	74.34	75.06

7 Conclusions

This study examined the performance of Transformer-based SER models on Italian data, focusing specifically on dataset preparation, model fine-tuning, and cross-corpus evaluation.

Through more strict selection criteria and targeted augmentation to balance emotional classes, an improvement in performance by up to 13% compared to the former method using the same base architecture [18] was attained. On our in-domain test set, the model achieved 93.56% accuracy, showing the usefulness of painstakingly collected Italian speech data.

When evaluated on external datasets, however, like EMOVO, accuracy fell to 61.73%, testifying to the notorious domain mismatch problem. Cross-corpus testing on alternative datasets revealed our model to still be competitive, beating its base model and holding the performance of larger models, but without the computational expense of models such as Whisper large V3.

These results underscore the importance of careful data preparation in closing the gap between experimental control and real-world deployment. By illustrating how performance varies between datasets, we have defined a framework for improving generalizability and mitigating the degradation characteristic of cross-corpus environments. In particular, the combination of data filtering, class-balanced representation, and targeted augmentations was determined to be instrumental in improving both in-domain accuracy and robustness to novel corpora.

Building on this foundation, future research can combine domain-adaptive methods and multimodal data to continue to enhance the robustness of SER in various clinical and nonclinical settings, continuing to move us toward an ultimately generalizable Speech Emotion Recognition algorithm.

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Fai una domanda



Cerca



Avvia il ragionamento



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